
VECTOR SPACE PROBLEM SHEET

1

- (a) Show that if the columns, $\mathbf{m}_j$, of an $n \times n$ matrix, $\mathbf{M}$, are not independent then $\mathbf{M}$ does not have a unique inverse. For the columns to not be independent implies there exists some set of coefficients c_j such that

$$\sum_{j=1}^n c_j \mathbf{m}_j = \mathbf{0}.$$

We note that the i^{th} element of $\mathbf{m}_j = M_{ij}$ so

$$\sum_{j=1}^n c_j M_{ij} = 0$$

We consider the vector $\mathbf{c}$ with elements c_i , then the i^{th} component of $\mathbf{M}\mathbf{c}$ is equal to

$$[\mathbf{M}\mathbf{c}]_i = \sum_{j=1}^n M_{ij} c_j = 0$$

or $\mathbf{M}\mathbf{c} = \mathbf{0}$. Consequently for any vector $\mathbf{x}$ and any scalar b we have that

$$\mathbf{M}(\mathbf{x} + b\mathbf{c}) = \mathbf{M}\mathbf{x}$$

if we denote $\mathbf{y} = \mathbf{M}\mathbf{x}$, then there is not a unique vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{M}\mathbf{x}$, but rather a family of vectors $\mathbf{x} + b\mathbf{c}$ for any b .

[3 marks]

- (b) Show that for three $n \times n$ matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ the inverse of $\mathbf{ABC}$ is $\mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$. You can do this by multiplying the product of matrices, but its inverse and use associativity of matrix multiplication

$$\begin{aligned} (\mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{ABC}) &= \mathbf{C}^{-1}\mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{BC} \\ &= \mathbf{C}^{-1}(\mathbf{B}^{-1}\mathbf{B})\mathbf{C} = \mathbf{C}^{-1}\mathbf{C} = \mathbf{I} \end{aligned}$$

[3 marks]

- (c) For non-square matrices, $\mathbf{A}$ and $\mathbf{B}$ we can have $\mathbf{AB} = \mathbf{I}$ while $\mathbf{BA} \neq \mathbf{I}$. Show this is the case for the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \end{pmatrix} = \mathbf{I}_1$$

$$\mathbf{BA} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

[3 marks]

- (d) For finite-dimensional square matrices if $\mathbf{AB} = \mathbf{I}$ then $\mathbf{BA} = \mathbf{I}$. Thus $\mathbf{B} = \mathbf{A}^{-1}$. This is not entirely trivial to prove. We break the proof down. Show that if $\mathbf{AB} = \mathbf{I}$ then $\mathbf{B}$ is injective (one-to-one). This means that for two different vectors $\mathbf{x}$ and $\mathbf{y}$ then $\mathbf{Bx} \neq \mathbf{By}$. Equivalently $\mathbf{B}(\mathbf{x} - \mathbf{y}) \neq \mathbf{0}$, unless $\mathbf{x} = \mathbf{y}$, but as $\mathbf{x}$ and $\mathbf{y}$ are arbitrary the only vector for which $\mathbf{Bx} = \mathbf{0}$ is the vector $\mathbf{x} = \mathbf{0}$. Prove that if $\mathbf{AB} = \mathbf{I}$ and $\mathbf{Bx} = \mathbf{0}$ then $\mathbf{x} = \mathbf{0}$.

Suppose $\mathbf{Bx} = \mathbf{0}$ then multiply by $\mathbf{A}$

$$\mathbf{ABx} = \mathbf{0}$$

$$\mathbf{Ix} = \mathbf{0}$$

Thus $\mathbf{x} = \mathbf{0}$.

[2 marks]

- (e) Next we want to show that in a finite dimensional space a square matrix that is injective is also surjective (onto). That is, if $\mathbf{B}$ is injective then for any vector $\mathbf{y}$ there exists a vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{B}\mathbf{x}$. The reason a finite dimensional square matrix that is injective is also surjective is not so obvious. To prove this we consider a set of n **basis** vectors that span the entire space $\mathbb{R}^n$. We call these $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. Because these are independent the only set of numbers c_i such that $\sum_i c_i \mathbf{v}_i = \mathbf{0}$ are when $c_1 = c_2 = \dots = c_n = 0$. Show that the set of vectors $\mathbf{v}'_i = \mathbf{B}\mathbf{v}_i$ form a set of basis vectors (i.e. they are linear independent). This is easily proved by contradiction.

Since $\mathbf{B}$ is injective and $\mathbf{v}_i \neq \mathbf{0}$ (since basis vectors are non-zero), clearly $\mathbf{v}'_i = \mathbf{B}\mathbf{v}_i$ are non-zero. The non-trivial part is to show these vectors are linear independent. Suppose this is not the case and there exists a set of coefficients c'_i such that

$$\begin{aligned}\sum_{i=1}^n c'_i \mathbf{v}'_i &= \mathbf{0} \\ \sum_{i=1}^n c'_i \mathbf{B}\mathbf{v}_i &= \mathbf{0} \\ \mathbf{B} \sum_{i=1}^n c'_i \mathbf{v}_i &= \mathbf{0}.\end{aligned}$$

But, as $\mathbf{B}$ is injective this implies

$$\sum_{i=1}^n c'_i \mathbf{v}_i = \mathbf{0}.$$

Since the vectors $\mathbf{v}_i$ are linear independent, so this is a contradiction.

[5 marks]

- (f) Use the fact that the vectors $\mathbf{v}'_i$ are linearly independent to show that, for any vector $\mathbf{y}$, we can find a vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{B}\mathbf{x}$

For any vector $\mathbf{y}$ can be represented by a weighted sum of n independent vectors. In particular we can write

$$\begin{aligned}\mathbf{y} &= \sum_{i=1}^n c_i \mathbf{v}'_i = \sum_{i=1}^n c_i \mathbf{B}\mathbf{v}_i \\ &= \mathbf{B} \left(\sum_{i=1}^n c_i \mathbf{v}_i \right) = \mathbf{B}\mathbf{x}\end{aligned}$$

where $\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i$.

[2 marks]

(g) Using the fact that $\mathbf{AB} = \mathbf{I}$ and $\mathbf{B}$ is surjective show $\mathbf{BA} = \mathbf{I}$

We note that for all x

$$\begin{aligned}\mathbf{AB}x &= x \\ \mathbf{BAB}x &= \mathbf{B}x\end{aligned}$$

But as $\mathbf{B}$ is surjective for any vector y there exists a vector x such that $y = \mathbf{B}x$. Thus,

$$\mathbf{BA}y = y,$$

but as this is true for any vector y it follows that $\mathbf{BA} = \mathbf{I}$.

[2 marks]

End of question 1

2 Throughout this question you may use two standard properties of the trace, namely that a matrix and its transpose have the same trace, and that the trace is unchanged by cyclically permuting a product

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^T)$$

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA}).$$

(a) For a vector $\mathbf{a} \in \mathbb{R}^m$, a vector $\mathbf{b} \in \mathbb{R}^n$ and an $m \times n$ matrix $\mathbf{U}$, show that

$$\mathbf{a}^T \mathbf{U} \mathbf{b} = \text{tr}(\mathbf{U}^T \mathbf{a} \mathbf{b}^T).$$

The quickest route is to write both sides out in components. Note that $\mathbf{a} \mathbf{b}^T$ is an $m \times n$ matrix while $\mathbf{a}^T \mathbf{U} \mathbf{b}$ is a scalar.

Writing the left-hand side in components

$$\mathbf{a}^T \mathbf{U} \mathbf{b} = \sum_{i=1}^m \sum_{j=1}^n a_i U_{ij} b_j.$$

For the right-hand side, the outer product $\mathbf{a} \mathbf{b}^T$ has elements $[\mathbf{a} \mathbf{b}^T]_{ik} = a_i b_k$, so

$$[\mathbf{U}^T \mathbf{a} \mathbf{b}^T]_{jk} = \sum_{i=1}^m [\mathbf{U}^T]_{ji} [\mathbf{a} \mathbf{b}^T]_{ik} = \sum_{i=1}^m U_{ij} a_i b_k.$$

The trace is the sum of the diagonal elements, $k = j$, so

$$\text{tr}(\mathbf{U}^T \mathbf{a} \mathbf{b}^T) = \sum_{j=1}^n \sum_{i=1}^m U_{ij} a_i b_j = \sum_{i=1}^m \sum_{j=1}^n a_i U_{ij} b_j,$$

which is the left-hand side.

Alternatively, $\mathbf{a}^T \mathbf{U} \mathbf{b}$ is a scalar, so it equals its own trace, and using $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ together with $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^T)$ gives the same result.

[5 marks]

(b) Verify the identity explicitly for

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \mathbf{U} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

by computing both $\mathbf{a}^\top \mathbf{U} \mathbf{b}$ and $\text{tr}(\mathbf{U}^\top \mathbf{a} \mathbf{b}^\top)$.

For the left-hand side,

$$\mathbf{U} \mathbf{b} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 13 \end{pmatrix} \quad \mathbf{a}^\top \mathbf{U} \mathbf{b} = (1 \quad 2) \begin{pmatrix} 5 \\ 13 \end{pmatrix} = 5 + 26 = 31.$$

For the right-hand side, the outer product is

$$\mathbf{a} \mathbf{b}^\top = \begin{pmatrix} 1 \\ 2 \end{pmatrix} (3 \quad 1) = \begin{pmatrix} 3 & 1 \\ 6 & 2 \end{pmatrix}$$

and hence

$$\mathbf{U}^\top \mathbf{a} \mathbf{b}^\top = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 6 & 2 \end{pmatrix} = \begin{pmatrix} 21 & 7 \\ 30 & 10 \end{pmatrix}$$

whose trace is $21 + 10 = 31$, agreeing with the left-hand side.

Note that only the diagonal elements are needed, so the off-diagonal entries 7 and 30 need never be computed. Note also that $\mathbf{U}$ is not symmetric: had the transpose been omitted the two sides would not have agreed, which is the point of checking with a non-symmetric matrix.

[4 marks]

- (c) For a square matrix $\mathbf{U}$ and a sufficiently small scalar ϵ , the inverse of $\mathbf{I} + \epsilon\mathbf{U}$ is given by the geometric (Neumann) series

$$(\mathbf{I} + \epsilon\mathbf{U})^{-1} = \mathbf{I} - \epsilon\mathbf{U} + \epsilon^2\mathbf{U}^2 - \epsilon^3\mathbf{U}^3 + \dots$$

Show this by multiplying $\mathbf{I} + \epsilon\mathbf{U}$ by the truncated series

$$\mathbf{S}_n = \sum_{k=0}^n (-\epsilon)^k \mathbf{U}^k$$

and showing that the result differs from $\mathbf{I}$ only at order ϵ^{n+1} . Hence write down $(\mathbf{I} + \epsilon\mathbf{U})^{-1}$ to first order in ϵ .

Multiplying out, and using $\epsilon(-\epsilon)^k = -(-\epsilon)^{k+1}$,

$$\begin{aligned} (\mathbf{I} + \epsilon\mathbf{U})\mathbf{S}_n &= \sum_{k=0}^n (-\epsilon)^k \mathbf{U}^k + \epsilon\mathbf{U} \sum_{k=0}^n (-\epsilon)^k \mathbf{U}^k \\ &= \sum_{k=0}^n (-\epsilon)^k \mathbf{U}^k - \sum_{k=0}^n (-\epsilon)^{k+1} \mathbf{U}^{k+1} \\ &= \sum_{k=0}^n (-\epsilon)^k \mathbf{U}^k - \sum_{j=1}^{n+1} (-\epsilon)^j \mathbf{U}^j. \end{aligned}$$

The two sums are identical except for the $k = 0$ term of the first and the $j = n + 1$ term of the second, so everything between telescopes and

$$(\mathbf{I} + \epsilon\mathbf{U})\mathbf{S}_n = \mathbf{I} - (-\epsilon)^{n+1} \mathbf{U}^{n+1}.$$

The correction is of order ϵ^{n+1} , so $\mathbf{S}_n$ is the inverse up to that order, and taking $n \rightarrow \infty$ gives the full series provided ϵ is small enough that $\epsilon^n \mathbf{U}^n \rightarrow \mathbf{0}$ (it suffices that $|\epsilon|$ is smaller than the reciprocal of the largest magnitude eigenvalue of $\mathbf{U}$).

Keeping only the first two terms,

$$(\mathbf{I} + \epsilon\mathbf{U})^{-1} = \mathbf{I} - \epsilon\mathbf{U} + O(\epsilon^2).$$

[5 marks]

- (d) Use this to find the derivative of $g(\mathbf{X}) = \mathbf{a}^\top \mathbf{X}^{-1} \mathbf{b}$ with respect to the matrix $\mathbf{X}$. Replace $\mathbf{X}$ by $\mathbf{X} + \epsilon \mathbf{U}$, expand to first order in ϵ , and use the definition of the derivative with respect to a matrix

$$\lim_{\epsilon \rightarrow 0} \frac{g(\mathbf{X} + \epsilon \mathbf{U}) - g(\mathbf{X})}{\epsilon} = \text{tr} \left(\mathbf{U}^\top \frac{\partial g(\mathbf{X})}{\partial \mathbf{X}} \right)$$

which holds for every $\mathbf{U}$. Part (a) is what lets you read the derivative off.

Factorising $\mathbf{X}$ out on the left, $\mathbf{X} + \epsilon \mathbf{U} = \mathbf{X}(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U})$, so by the previous part

$$\begin{aligned} (\mathbf{X} + \epsilon \mathbf{U})^{-1} &= (\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U})^{-1} \mathbf{X}^{-1} = (\mathbf{I} - \epsilon \mathbf{X}^{-1} \mathbf{U} + O(\epsilon^2)) \mathbf{X}^{-1} \\ &= \mathbf{X}^{-1} - \epsilon \mathbf{X}^{-1} \mathbf{U} \mathbf{X}^{-1} + O(\epsilon^2). \end{aligned}$$

Therefore

$$g(\mathbf{X} + \epsilon \mathbf{U}) - g(\mathbf{X}) = -\epsilon \mathbf{a}^\top \mathbf{X}^{-1} \mathbf{U} \mathbf{X}^{-1} \mathbf{b} + O(\epsilon^2)$$

and dividing by ϵ and letting $\epsilon \rightarrow 0$

$$\lim_{\epsilon \rightarrow 0} \frac{g(\mathbf{X} + \epsilon \mathbf{U}) - g(\mathbf{X})}{\epsilon} = -\mathbf{a}^\top \mathbf{X}^{-1} \mathbf{U} \mathbf{X}^{-1} \mathbf{b}.$$

This is of the form $\mathbf{c}^\top \mathbf{U} \mathbf{d}$ with $\mathbf{c}^\top = -\mathbf{a}^\top \mathbf{X}^{-1}$, that is $\mathbf{c} = -(\mathbf{X}^{-1})^\top \mathbf{a}$, and $\mathbf{d} = \mathbf{X}^{-1} \mathbf{b}$. By part (a),

$$\mathbf{c}^\top \mathbf{U} \mathbf{d} = \text{tr}(\mathbf{U}^\top \mathbf{c} \mathbf{d}^\top).$$

Comparing with the definition above, and since this holds for every $\mathbf{U}$, the derivative is

$$\frac{\partial g(\mathbf{X})}{\partial \mathbf{X}} = \mathbf{c} \mathbf{d}^\top = -(\mathbf{X}^{-1})^\top \mathbf{a} \mathbf{b}^\top (\mathbf{X}^{-1})^\top.$$

As a check on the shapes, $\mathbf{a} \mathbf{b}^\top$ is a matrix of the same size as $\mathbf{X}$, as the derivative must be.

[6 marks]

End of question 2

3 We want to compute the derivative of $\log(\det(\mathbf{X}))$ with respect to the matrix $\mathbf{X}$. Our strategy will be to use the fact that

$$\lim_{\epsilon \rightarrow 0} \frac{\log(\det(\mathbf{X} + \epsilon \mathbf{U})) - \log(\det(\mathbf{X}))}{\epsilon} = \text{tr} \mathbf{U}^T \frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}.$$

We therefore expand $\log(\det(\mathbf{X} + \epsilon \mathbf{U}))$ to first order in ϵ , subtract $\log(\det(\mathbf{X}))$, divide by ϵ , rearrange and read off $\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}$. However, rearranging in this case is quite complicated.

- (a) Using $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ and $\log(ab) = \log(a) + \log(b)$ expand $\log(\det(\mathbf{X} + \epsilon \mathbf{U}))$ as a sum of $\log(\det(\mathbf{X}))$ and another logarithm.

$$\begin{aligned} \log(\det(\mathbf{X} + \epsilon \mathbf{U})) &= \log(\det(\mathbf{X}(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U}))) \\ &= \log(\det(\mathbf{X}) \det(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U})) \\ &= \log(\det(\mathbf{X})) + \log(\det(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U})) \end{aligned}$$

[5 marks]

- (b) The hard part is to expand $\det(\mathbf{I} + \epsilon \mathbf{M})$. Write out $\det(\mathbf{I} + \epsilon \mathbf{M})$ explicitly for a 2 dimensional matrix and expand to first order in ϵ

$$\begin{aligned} \det(\mathbf{I} + \epsilon \mathbf{M}) &= \det \begin{pmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} \\ \epsilon M_{21} & 1 + \epsilon M_{22} \end{pmatrix} = (1 + \epsilon M_{11})(1 + \epsilon M_{22}) - \epsilon^2 M_{21} M_{12} \\ &= 1 + \epsilon(M_{11} + M_{22}) + O(\epsilon^2) \end{aligned}$$

[2 marks]

(c) For an $n \times n$ matrix $\mathbf{M}$ compute the determinant of

$$\det(\mathbf{I} + \epsilon \mathbf{M}) = \det \begin{pmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} & \dots & \epsilon M_{1n} \\ \epsilon M_{21} & 1 + \epsilon M_{22} & \dots & \epsilon M_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon M_{n,1} & \epsilon M_{n2} & \dots & 1 + \epsilon M_{nn} \end{pmatrix}$$

up to first order in ϵ . Recall we can compute the determinant a matrix $\mathbf{X}$ as

$$\det(\mathbf{X}) = \sum_{i=1}^n (-1)^j X_{1j} \det(\hat{\mathbf{X}}_{1j})$$

where $\hat{\mathbf{X}}_{1j}$ is the matrix obtained by removing row 1 and column j from the matrix $\mathbf{X}$.

Expanding the determinant we get a term equal to the product of the diagonal elements

$$\prod_{i=1}^n (1 + \epsilon M_{ii}) = 1 + \epsilon \sum_{i=1}^n M_{ii} + O(\epsilon^2)$$

all other terms in the determinant involve at least terms of order ϵ^2 so do not contribute to order ϵ . In consequence

$$\det(\mathbf{I} + \epsilon \mathbf{M}) = 1 + \epsilon \text{tr} \mathbf{M} + O(\epsilon^2)$$

[5 marks]

(d) For a more elegant proof we can use the fact that

$$\det(\mathbf{M}) = \prod_{i=1}^n \lambda_i \quad \text{tr} \mathbf{M} = \sum_{i=1}^n \lambda_i$$

where λ_i are the eigenvalues of the matrix $\mathbf{M}$. We will prove these results later when we look at eigen-systems. Note that if $\mathbf{v}_i$ is an eigenvector of the matrix $\mathbf{M}$, with eigenvalue λ_i then it is an eigenvector of $\mathbf{I} + c\mathbf{M}$ with eigenvalue $1 + c\lambda_i$ since

$$(\mathbf{I} + c\mathbf{M})\mathbf{v}_i = \mathbf{v}_i + c\lambda_i \mathbf{v}_i = (1 + c\lambda_i)\mathbf{v}_i.$$

Using these results show that $\det(\mathbf{I} + \epsilon \mathbf{M}) = 1 + \epsilon \text{tr} \mathbf{M} + O(\epsilon^2)$.

$$\begin{aligned} \det(\mathbf{I} + \epsilon \mathbf{M}) &= \prod_{i=1}^n (1 + \epsilon \lambda_i) = 1 + \epsilon \sum_{i=1}^n \lambda_i + O(\epsilon^2) \\ &= 1 + \epsilon \text{tr} \mathbf{M} + O(\epsilon^2) \end{aligned}$$

[3 marks]

(e) Using the results

$$\begin{aligned}\log(\det(\mathbf{X} + \epsilon \mathbf{U})) - \log(\det(\mathbf{X})) &= \log(\det(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U})) \\ \det(\mathbf{I} + \epsilon \mathbf{M}) &= 1 + \epsilon \text{tr} \mathbf{M} + O(\epsilon^2)\end{aligned}$$

together with the fact that $\log(1 + x) = x + O(x^2)$, compute $\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}$. You may use the fact that $\log(1 + x) = x + O(x^2)$ and $\text{tr} \mathbf{A} = \text{tr} \mathbf{A}^\top$.

We note that

$$\begin{aligned}\log(\det(\mathbf{X} + \epsilon \mathbf{U})) - \log(\det(\mathbf{X})) &= \log(1 + \epsilon \text{tr} \mathbf{X}^{-1} \mathbf{U} + O(\epsilon^2)) \\ &= \epsilon \text{tr} \mathbf{X}^{-1} \mathbf{U} + O(\epsilon^2) \\ &= \epsilon \text{tr} \mathbf{U}^\top (\mathbf{X}^{-1})^\top\end{aligned}$$

Thus,

$$\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}} = (\mathbf{X}^{-1})^\top$$

[5 marks]

End of question 3
END OF PAPER